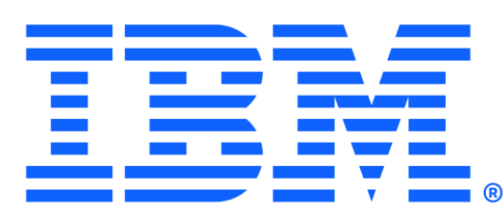


Fine-Tuned Thoughts: Leveraging Chain-of-Thought Reasoning for Industrial Asset Health Monitoring

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Motivation: Knowledge in Industry

IoT sensors → capture signals. **Failure Mode and Effects Analysis (FMEA)** links failures ↔ sensors.

Goal: predict failures when anormal sensor behavior is detected. Or, what sensor should be monitored to detect a failure early?

Challenges:

- Lack of data makes reasoning over heterogeneous industrial assets hard.
- Bringing Large LM reasoning

to resource-constrained industrial environments.

- Small LMs weaker in complex reasoning vs LLMs.

Contributions:

- A KnowledgeGraph-inspired knowledge distillation pipeline transferring CoT reasoning from LLMs to SLMs on MCQA tasks for asset health monitoring
- A qualitative evaluation of in-context learning and supervised fine-tuning using generated knowledge

Expert Knowledge: Failure Faults ↔ Sensors/Parameters

Failure Mode	Sensor/Parameter Reading				
	Power	Speed	Pressure	Vibr.	Temp.
Bearing wear		✓	✓		✓
Gear Defect			✓	✓	
Unbalance	✓				✓
Shaft Misalignment	✓	✓		✓	
Overheating			✓		✓

How Good is Generated Knowledge?

Coverage: 54 assets, 66 sensors, and 59 failure modes from ISO standards.

Data Stats: 6.2k, 6.1k, and 6.2k samples for CoT-Expert, CoT-Inductive, and CoT-Standard.

Factuality Analysis:

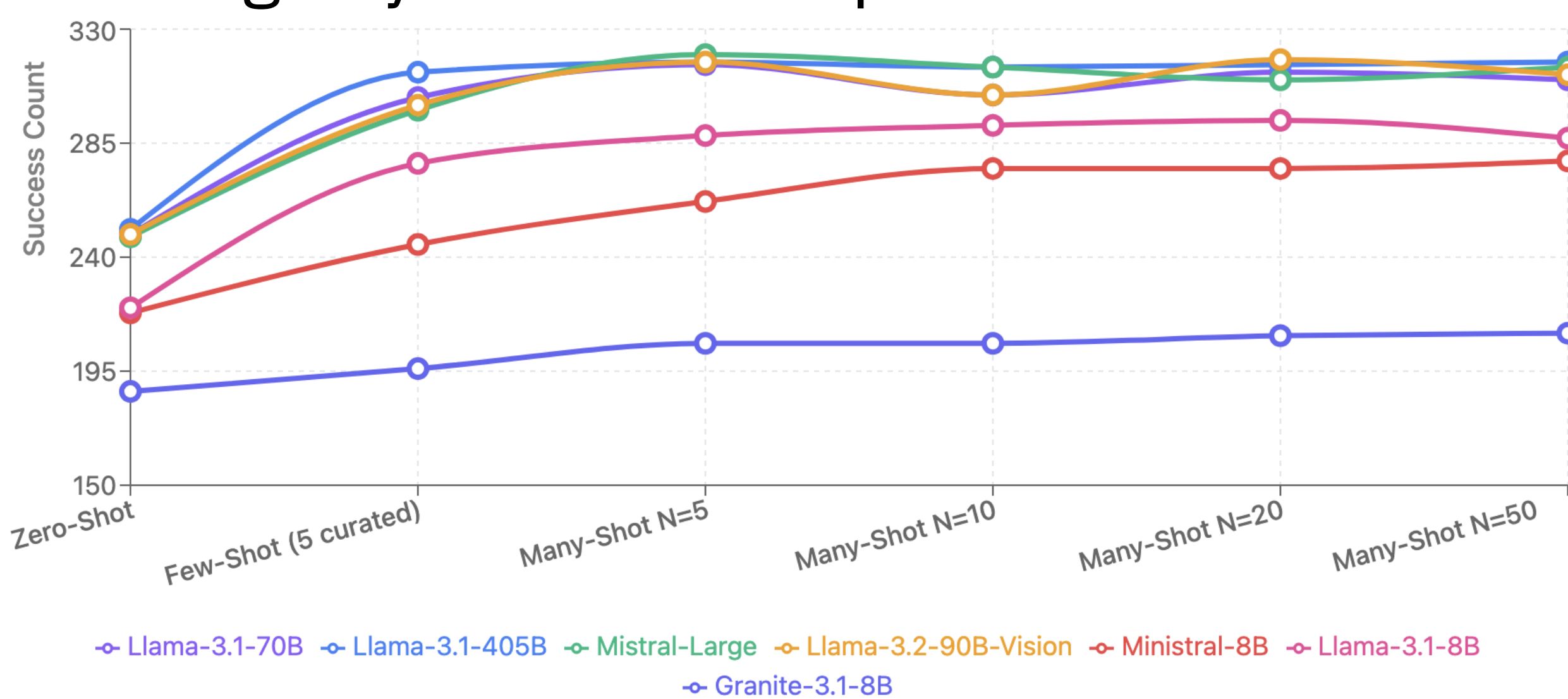
- Evaluated by FActScore (EMNLP 2023)
- nearly 70% of facts of our teacher LLM generations are backed by trusted source

	Responding (%)	FActScore (%)
FailureSensorIQ	94.7	69.8
Generated data	89.6	70.8

Factuality Comparison Between FailureSensorIQ and Teacher Generated Data

Use of Generated Knowledge: ICL & SFT

In-Context Learning: MiniLM-L6-v2 embedding-based retrieved examples improved model accuracy, most notably for larger models, showing substantial gains from zero-shot to few-shot setups, with performance plateauing beyond 20 examples.



QLoRA Fine-Tuning: boosts accuracy across models, enabling smaller models to approach larger-model performance, while CoT fine-tuning makes direct prompting most effective and reduces invalid outputs, though with a slight rise in confident wrong answers.

Model	Experiment Settings		Evaluation Scores				
	Synth. Data	Prompting	$P_{\text{single-correct}}$	P_{invalid}	$P_{\text{mul-correct}}$	$P_{\text{single-wrong}}$	$P_{\text{mul-wrong}}$
(baselines)							
Llama-3.1-405B-Instruct	N/A	CoT Std	0.5126	0.0019	0.1691	0.258	0.0585
Llama-3.1-8B-Instruct	N/A	direct	0.4012	0.012	0.1991	0.3048	0.0829
Mistral-Large-Instruct	N/A	direct	0.5009	0.0244	0.186	0.2295	0.0592
Ministral-8B-Instruct	N/A	direct	0.264	0	0.4113	0.1894	0.1354
Granite-3.1-8B Instruct	N/A	direct	0.2411	0.0015	0.4046	0.1916	0.1612
FT on Llama-3.1-8B-Instruct	$D_{\text{CoT-Std-gen}}$	direct	0.5111	0.0071	0.1095	0.3116	0.0607
	$D_{\text{CoT-Exp-gen}}$	direct	0.4698	0.0049	0.0979	0.375	0.0525
	$D_{\text{CoT-Ind-gen}}$	CoT Ind	0.4387	0.0165	0.1365	0.3168	0.0915
FT on Ministral-8B-Instruct	$D_{\text{CoT-Std-gen}}$	direct	0.4402	0	0.144	0.3573	0.0585
	$D_{\text{CoT-Exp-gen}}$	direct	0.4623	0.0004	0.1301	0.3495	0.0577
	$D_{\text{CoT-Ind-gen}}$	direct	0.4938	0	0.1537	0.2913	0.0611
FT on Granite-3.1-8B-Instruct	$D_{\text{CoT-Std-gen}}$	CoT Std	0.3813	0.0427	0.1552	0.3142	0.1065
	$D_{\text{CoT-Exp-gen}}$	direct	0.4083	0	0.2583	0.2163	0.117
	$D_{\text{CoT-Ind-gen}}$	CoT Std	0.4062	0.006	0.1407	0.3952	0.0519

Ablation Studies

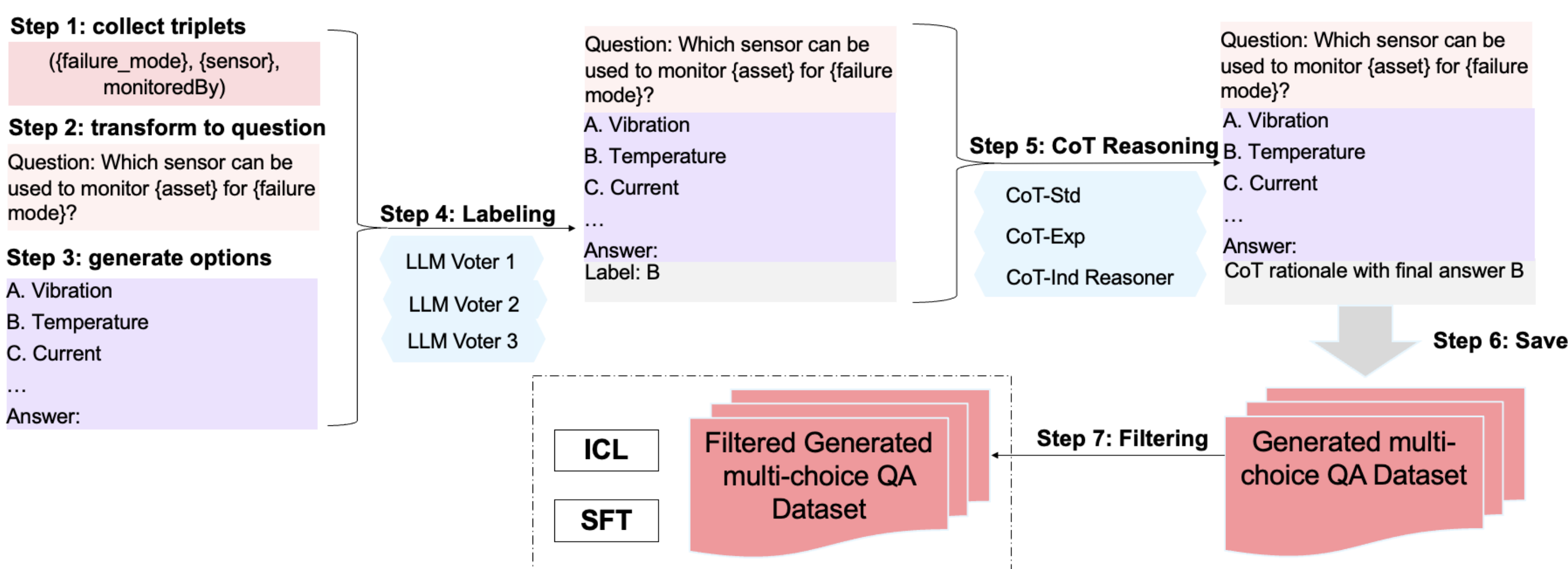
Reasoning Capability

- Without rationale (direct answer only)
- Incorrect pseudo-labeling (mismatched answer-rationale pairs)

Resilience to Perturbation

- Option ID Shifting
- Knowledge-Invariant Question Paraphrasing

Knowledge Distillation Pipeline



Knowledge Graph Structure

Instruction Generation

(s, o, r)
 s o r
subject object relation

Step 1. Omit subject (s) or object (o).
e.g. $T(_, [asset], mountedOn) \rightarrow$ "What sensors are mounted on [asset]?"
Step 2. Apply seed templates
Step 3. Generate LLM instructions

Options Generation

- **Correct Answers**: For each question Q , teacher LLMs rank options from a universal set U based on predefined correctness criteria; the top $K = 5$ options are treated as correct answers
- **Distractors**: The bottom $2K$ least relevant options from the ranking are used as distractor candidates, and answer positions are randomized to avoid predictable patterns.

Pseudo Ground Truth

Labelling

Majority voting with:

- Mixtral Large
- Llama 3.1 405b
- ChatGPT 4o

Rationale Generation

CoT Type	Trigger Statement
Standard	Let me think step by step
Inductive	Let me think step by step as a reliability engineer
Expert	Let's use step by step inductive reasoning, given the domain specific nature of the question

CoT Triggering Statement

Quality Filtering

FILTER 1: LENGTH	FILTER 2: DISTANCE	FILTER 3: DIFFICULTY	FILTER 4: QUALITY
Total characters (input + output)	Embedding similarity to nearest neighbor	LLM-as-Judge (5 scales)	LLM-as-Judge (5 scales)
		Very-Easy Easy Medium Hard Very Hard	Very-Poor Poor Average Good Excellent